

Pre-work • Nervous Tissue and Neuronal Integration

Functional Organization and Nervous Tissue, and Gross Anatomy and Neuronal Integration. Read both Part A chapters first. This packet does not repeat them. Every other packet in Module 5 is built on the cell and the vocabulary in these two chapters, so the time you spend here is repaid four more times.

Module 5 - 1 of 5

Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

- 1 Read**
Read the Part A chapter first. This packet does not repeat it. It puts it to work.
- 2 Organize**
Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
- 3 Draw**
Panel 2. Brain dump from memory, then draw, then correct in a second colour.
- 4 Apply**
Panel 3. Notes closed for the first pass. Open them only to check.
- 5 Retrieve**
Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the two tissue chapters into mini visuals. Eight chunks. The grids carry the naming pairs, the hubs carry the structure, and the one chain is the reflex arc.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 HUB

The nervous system and its divisions

Nervous system at the hub. Spoke first to the CNS with the brain and spinal cord, then to the PNS with nerves, ganglia, and sensory receptors. Off the motor side of the PNS hang somatic, autonomic, and enteric, and off autonomic hang sympathetic and parasympathetic. Write the effector each motor division reaches at the end of its spoke.

2 HUB

The neuron

One cell body in the middle. Spoke out to dendrites, axon hillock, axon with axoplasm and axolemma, and axon terminals. Inside the cell body mark the nucleus, the Nissl bodies, the neurofibrils, and the microtubules. Draw an arrow along the whole cell showing which way an impulse travels, because the direction is what names the parts.

3 GRID

Classifying neurons, twice

Neuron type down the side: multipolar, bipolar, unipolar. Number of processes, where it is found, and which functional class usually wears that shape across the top. Then add two rows at the bottom for Purkinje and pyramidal cells and write where each one lives.

4 GRID

The six neuroglia

Glial cell down the side, all six. CNS or PNS, and the job it does across the top. Add a third column: what goes wrong if this cell fails. Four rows are CNS and two are PNS, and keeping them grouped is half the learning.

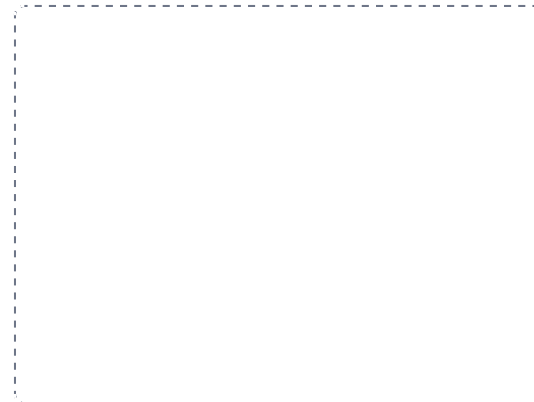
DRAWING 1

A multipolar neuron, fully labelled

BRAIN DUMP FIRST

Two minutes. Every part of the cell, and the direction a signal moves through it.

Draw one large multipolar neuron. Give it many branched dendrites, one cell body with the nucleus and Nissl bodies inside, a clear axon hillock, and one long axon ending in branched terminals with synaptic end bulbs. Wrap the axon in a myelin sheath with nodes of Ranvier between the segments. Now enlarge one end bulb in a bubble and draw the synaptic vesicles, the cleft, and the next cell's membrane.



In your notebook, head the page **M5-1 A multipolar neuron, fully labelled** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can put your finger on any part of your drawing and say whether a signal at that point is travelling toward the cell body or away from it.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A patient cuts a nerve in the forearm and, over many months, sensation and movement return. Another patient has a tract severed in the spinal cord and recovers nothing. Both injuries cut axons. Explain the difference from structure.

Answer. The forearm nerve is myelinated by Schwann cells, which leave a neurolemma. The cord tract is myelinated by oligodendrocytes, which do not.

Reasoning. Start with the cell that builds the sheath, not with the axon. In the PNS a Schwann cell wraps one segment of one axon, and its outer layer, the neurolemma, survives the injury as a hollow tube. That tube is a guide, so the regrowing axon has a path back to its target. In the CNS one oligodendrocyte reaches up to fifteen axons and leaves no neurolemma behind, so a regrowing axon has nothing to follow. The location of the injury also matters, because an axon cut off from its cell body loses its supply of proteins and dies beyond the cut. So the answer is not about the axon at all, it is about which glial cell was doing the wrapping.

Set A - Name it

1. The gaps in the myelin sheath along an axon are the _____.
2. A cluster of neuron cell bodies located outside the CNS is a _____.
3. The glial cell that lines the ventricles and helps circulate cerebrospinal fluid is the _____.
4. The cone-shaped region where the cell body joins the axon is the _____.
5. Name the five components of a reflex arc, in order.

5 SPLIT

PNS myelination against CNS myelination

One split. Which cell builds the sheath, how many axons one cell can reach, neurolemma present or absent, and whether a cut axon can regrow. This is the pair that answers the regeneration question, so make the neurolemma row impossible to miss.

6 SPLIT

Gray matter against white matter

One split. What each is made of, and where each sits. Then hang two small sketches off the branches: a spinal cord cross-section with gray inside, and a brain section with gray outside. The swap between the two organs is the whole point of the visual.

7 GRID

Tract, nerve, nucleus, ganglion

A true two by two. Bundle of axons and cluster of cell bodies down the side, CNS and PNS across the top. Write the term in each of the four cells, then add one example of each. If you can fill this grid blank you have four terms for the price of one.

8 CHAIN

The reflex arc, five components

Five boxes with arrows, receptor through to effector. Write beside each box where that component physically sits, and mark the one box that is inside the CNS. Then note beside the integration center whether the arc is monosynaptic or polysynaptic.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

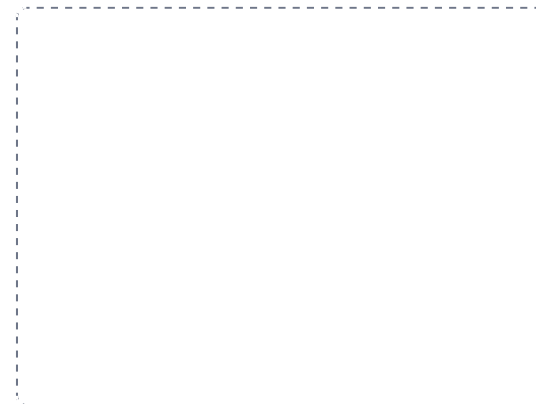
DRAWING 2

A myelinated axon in the PNS and in the CNS, side by side

BRAIN DUMP FIRST

One minute. Which cell wraps each one, and what is left behind on the outside.

Draw two axons beside each other. On the left, wrap the PNS axon with Schwann cells, one per segment, and draw the neurolemma as the outer layer of each Schwann cell. On the right, draw one oligodendrocyte with several arms, each arm wrapping a segment of a different CNS axon, and leave the outer surface bare. Mark the nodes of Ranvier on both and draw the impulse jumping node to node.



In your notebook, head the page **M5-2 A myelinated axon in the PNS and in the CNS, side by side** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain from your two drawings why a cut nerve in the forearm can regrow and a cut tract in the cord usually cannot.

Set B · Predict it

1. The immune system strips myelin from axons in the CNS. Predict the effect on impulse speed and name the cell that was lost.

2. An axon is cut halfway along its length. Predict what happens to the portion beyond the cut and explain why, using the cell body.

3. A neuronal pool is wired for divergence. Predict what one incoming signal does to the number of neurons responding.

4. A stroke damages a descending pathway on the right side of the brain. Predict which side of the body is weakened and name the reason.

Set C · Tell it apart

1. Distinguish a tract from a nerve, and a nucleus from a ganglion, by what each is made of and where it sits.

2. Explain why a tumor of nervous tissue almost always arises from glia rather than from neurons.

3. Distinguish a monosynaptic from a polysynaptic reflex by the wiring in the integration center, and give an example of each.

DRAWING 3

Two reflexes drawn as arcs

BRAIN DUMP FIRST

Two minutes. All five components in each, and the difference in the middle.

Draw a spinal cord cross-section twice. On the first, draw the patellar reflex: the muscle spindle receptor, the sensory neuron entering the posterior root, one synapse in the cord, the motor neuron leaving the anterior root, and the quadriceps as the effector. On the second, draw the withdrawal reflex with an interneuron added between the sensory and motor neurons. Label all five components on both and mark the posterior root ganglion.



In your notebook, head the page **M5-3 Two reflexes drawn as arcs** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can count the synapses on each of your drawings and name the reflex type from the count alone.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 The nervous system and its divisions

HUB

2 The neuron

HUB

3 Classifying neurons, twice

GRID

4 The six neuroglia

GRID

5 PNS myelination against CNS myelination

SPLIT

6 Gray matter against white matter

SPLIT

7 Tract, nerve, nucleus, ganglion

GRID

8 The reflex arc, five components

CHAIN